

Supplementary File 3

1. Purpose of This Document

This guide walks you through installing FoxWatch, loading your data, and reproducing the fox abundance results reported in the manuscript. It is written for users who are not R programmers. By the end you will be able to run the full FoxWatch workflow on your own camera-trap data.

The guide covers four steps:

- Installing the software FoxWatch requires
- Downloading and launching FoxWatch
- Reproducing the Oolambeyan National Park case study step by step
- Interpreting the results

2. System Requirements

FoxWatch runs on any standard desktop or laptop computer meeting the following minimum specifications:

Component	Minimum	Recommended
Operating system	Windows 10, macOS 12, Ubuntu 20.04	Windows 11, macOS 14, Ubuntu 22.04
Processor	Dual-core 2.0 GHz	Intel Core i7 quad-core or equivalent
RAM	8 GB	16 GB
Disk space	1 GB free	2 GB free
Internet	Required for installation	—
R version	≥ 4.2.0	4.5.2 (2025-10-31 ucrt)
JAGS version	≥ 4.3.1	4.3.1

Note: The Oolambeyan National Park case study (five years, 35 cameras, approximately 165 ten-day blocks) completes in approximately five minutes on an Intel Core i7 with 16 GB RAM.

3. Installation

3.1 Install R

1. Open a web browser and go to <https://cran.r-project.org>
2. Click the download link for your operating system.
3. Windows: click “base”, then download and run the installer with default settings.
4. macOS: download the .pkg file (use the arm64 build for M1/M2/M3 Macs) and double-click to install.
5. Linux (Ubuntu): follow the CRAN instructions — typically two or three terminal commands.

3.2 Install RStudio (free)

6. Go to <https://posit.co/download/rstudio-desktop/>
7. Download the installer for your operating system and run it with default settings.
8. Open RStudio to confirm it launches — you should see a four-panel window.

3.3 Install JAGS

JAGS (Just Another Gibbs Sampler) is the Bayesian engine FoxWatch uses to fit the abundance model. It must be installed as a separate program before FoxWatch can run the abundance model.

Windows: Go to <https://sourceforge.net/projects/mcmc-jags/files/JAGS/4.x/>, click the latest 4.x folder, then the Windows folder. Download the .exe installer and run it with default settings. Restart RStudio after installation.

macOS: Go to the same link, click the latest 4.x folder, then Mac OS X. Download and install the .pkg file.

Linux (Ubuntu): Open a terminal and run:

```
sudo apt update && sudo apt install jags
```

To confirm R can find JAGS, paste the following into the RStudio Console and press Enter:

```
install.packages("rjags")
library(rjags)
cat("JAGS version:", rjags::jags.version(), "\n")
```

You should see: JAGS version: 4.3.1. If you see an error, see Section 7 (Troubleshooting).

3.4 Install Required R Packages

Copy the block below, paste it into the RStudio Console, and press Enter. This installs all packages FoxWatch needs and only needs to be done once.

```
pkgs <- c(
  "shiny", "dplyr", "readr", "stringr", "DT", "tools",
  "ggplot2", "plotly", "shinycssloaders", "lubridate",
  "rjags", "modeest", "coda",
  "sf", "ggspatial", "viridis", "ggrepel", "hms",
  "colorspace", "RColorBrewer", "forcats"
)
to_install <- setdiff(pkgs, rownames(installed.packages()))
if (length(to_install)) install.packages(to_install)
invisible(lapply(pkgs, library, character.only = TRUE))
cat("All FoxWatch packages loaded successfully.\n")
```

Note: If a popup asks “Do you want to install from sources the package which needs compilation?”, click No.

4. Download and Launch FoxWatch

9. Go to the FoxWatch GitHub repository: [GitHub URL]
10. Click the green Code button, then Download ZIP.
11. Unzip the download to a convenient folder, for example Documents/FoxWatch.
12. You should see at minimum the file FoxWatch_Version1.R and an example_data/ folder containing Supplementary Files 1 and 2.
13. Open RStudio, go to File → Open File... and open FoxWatch_Version1.R.
14. Click the Run App button at the top right of the editor panel.

FoxWatch will open in a new window showing a black banner with the title “FoxWatch: Smart Data Management & Abundance Estimation” and three tabs: Spatial Analysis, Camera Data & Detection, and Fox Abundance Model.

5. Worked Example — Oolambeyan National Park

The following walkthrough reproduces the results reported in the manuscript using the example files provided in the repository. These files are derived from the same dataset used in Armstrong & McSorley (2024).

Part A — Spatial Analysis

This tab produces a map of fox detections per camera station per year.

Files needed:

- Supplementary_File_1.csv — fox detection records (columns: SiteID, Date, Time, Detection)
- Site locations CSV — one row per camera station (columns: SiteID, latitude, longitude)
- Park boundary shapefile — or a zip file containing the .shp, .shx, .dbf and .prj files

Steps:

15. In the left sidebar under Spatial Data, click Browse next to Fox Detection Records (CSV) and select Supplementary File 1.
16. Click Browse next to Site Locations (CSV) and select your site locations file.
17. Click Browse next to Park Boundary and select the Oolambeyan shapefile.
18. In the Spatial Analysis tab, set Minimum year to 2023 and leave Exclude year blank.
19. Click Generate Map.

Expected output: A strip of panels — one per year plus a Site Locations reference panel — with size-scaled coloured dots at each camera station. Larger dots indicate more detections. A scale bar appears bottom-left and a north arrow top-left. Click Download Map (PNG) to save a 300 dpi figure.

Part B — Camera Data & Detection

This tab reads raw camera-trap text files, extracts fox detection records automatically, and produces summary tables and charts.

Files needed: One or more camera text files in EXIF format. Supplementary File 2 is an example file for camera Ool_10.

Steps:

20. In the left sidebar under Camera Data, click Browse next to Camera TXT Files (EXIF) and select your camera files. Multiple files can be selected at once.
21. FoxWatch automatically creates a tab for each camera (e.g., Ool_10).
22. Click the tab for the camera you want to review. Parsed records appear in a searchable table showing Date, Time, Detection, and Image ID.
23. Check the table looks correct, then click Process Ool_10. A green notification confirms success.
24. Repeat Steps 3–4 for each camera.
25. Scroll down to the Summary Statistics table to see total fox detections per camera station.
26. The Detection Visualisation chart shows detections as an interactive bar chart.
27. Click Download Fox Detections to export processed records as a text file.

Note: Camera files must be named with the pattern `Ool_###` (for example, `Ool_10.txt`) for the camera ID to be extracted automatically. If your files use a different naming convention, FoxWatch will use the filename without its extension as the camera ID.

Part C — Fox Abundance Model

This tab fits a Bayesian Simultaneous-Count Model to a time series of daily fox detection counts and estimates fox abundance for each 10-day period.

Files needed: A plain text file with one integer count per line. Each line is the total number of fox detections across all cameras on that day.

Steps:

28. In the left sidebar under Abundance Data, click Browse next to Daily Detection Counts (TXT) and select your daily counts file.
29. Click the Fox Abundance Model tab.
30. Click Run Abundance Model.
31. A progress bar shows five stages: Building Bayesian SCM → Running MCMC chains → Sampling from posterior → Summarising results → Finalising.
32. When complete, the first ten rows of results appear in the preview table.
33. Scroll to the Abundance Visualisation chart to see the time series with shaded uncertainty ribbon.
34. Click Download Abundance Estimates to save all results as a CSV file.

The results table contains the following columns:

Column	Meaning
Period	The 10-day block number
Abundance	Posterior mode — the single most probable fox abundance estimate for that block
LowerHDI	Lower bound of the 95% highest posterior density (HPD) interval
UpperHDI	Upper bound of the 95% HPD interval
Rhat	Gelman–Rubin convergence statistic for the abundance parameter N in that block
Convergence	Traffic-light convergence label derived from Rhat (see Section 6)

Expected output for the Oolambeyan dataset: With the full five-year, 35-camera dataset you should see approximately 165 ten-day blocks. The abundance trend will show an initial high-abundance period followed by a sustained decline consistent with the results reported in Armstrong & McSorley (2024).

6. Understanding the Convergence Indicator

FoxWatch runs two independent MCMC chains for every 10-day block and checks whether they agree using the Gelman–Rubin R-hat statistic (Gelman & Rubin 1992). The results table and abundance plot use the following colour code:

Label	R-hat	Meaning
Good	≤ 1.05	Chains agree — estimate is reliable
Acceptable	1.05–1.10	Minor discrepancy — use with caution
Poor	> 1.10	Chains disagree — treat estimate with caution

What to do if a block shows Poor convergence:

- Check the daily detection counts for that period — very low or zero counts are the most common cause.
- If the problem persists across multiple blocks, review whether the survey period contains enough detections to support reliable estimation.
- Consult Armstrong & McSorley (2024) for guidance on minimum detection requirements.

7. Interpreting Your Results

Abundance estimate: The posterior mode — the single most probable number of foxes present during that 10-day block given the detection data.

95% HPD interval: This is a Bayesian credible interval, not a frequentist confidence interval. It means: given the model and the observed data, there is a 95% probability that the true fox abundance falls within this range. A narrow interval indicates a precise estimate; a wide interval indicates greater uncertainty, most commonly caused by sparse detection data during that period.

Three factors affect interval width, based on the simulation results of Armstrong & McSorley (2024):

- Survey duration is the most important factor — do not attempt to use blocks shorter than 10 days.
- Detection probability — low fox activity produces wide intervals.
- Camera number — adding more cameras has comparatively little effect on interval width relative to ensuring adequate survey duration.

Focus on trends, not individual blocks. Single-block estimates can be variable, particularly when fox abundance is low. The pattern across multiple consecutive blocks is more informative for management decisions than any single estimate.

8. Troubleshooting

Error or Symptom	Likely Cause and Fix
JAGS not found error when loading rjags	JAGS is not installed or R cannot find it. Re-install JAGS (Section 3.3) and restart RStudio. On Windows, confirm JAGS is on the system PATH.
package 'sf' failed to install (Linux)	Install system dependencies first: <code>sudo apt install libgdal-dev libproj-dev libgeos-dev libudunits2-dev</code> , then re-run <code>install.packages("sf")</code> .

Error or Symptom	Likely Cause and Fix
Map shows placeholder after uploading files	All three spatial files (Fox CSV, Sites CSV, Park shapefile) must be uploaded before clicking Generate Map. Check column names are exactly SiteID, latitude, longitude (lower-case for lat/lon).
Camera data table is empty after upload	The parser expects lines in the format IMG_###-dd/mm/yyyy hh:mm:ss AM/PM [...] -Fox. Open the file in a text editor to confirm the format.
"Need at least 10 daily counts" error	The daily counts file has fewer than 10 valid numeric lines. Open the file and check each line contains a single integer.
Abundance model run is very slow	Close other applications to free RAM. For datasets with more than 300 ten-day blocks, consider processing the data in annual segments.

9. Software Versions Used in Development

Component	Version	Notes
R	4.5.2 (2025-10-31 ucrt)	Minimum: 4.2.0
JAGS	4.3.1	Minimum: 4.3.1
Platform tested	Windows 11 / macOS 14 / Ubuntu 22.04	All three tested
shiny	1.8.0	UI framework
rjags	4-15	JAGS interface
coda	0.19-4	MCMC diagnostics and HPD intervals
modeest	2.4.0	Posterior mode estimation
ggplot2	3.5.1	Visualisation
plotly	4.10.4	Interactive plots
sf	1.0-16	Spatial data (Part A)
ggspatial	1.1.9	Map annotations
DT	0.33	Interactive tables
lubridate	1.9.3	Date and time parsing
readr	2.1.5	CSV and TSV input/output

10. Further Information

- For the full method description and scientific results, see the main manuscript.
- For the prior sensitivity discussion and model specification details, see Supplementary File 4.
- For questions, bug reports, or feature requests, use the Issues page of the FoxWatch GitHub repository or contact the corresponding author at dumie.dissanayake@dcceew.nsw.gov.au

References

Armstrong, G. & McSorley, A. 2024, Estimating fox abundance using the simultaneous count method, *Wildlife Letters*, vol. 2, pp. 102–109, doi:10.1002/wll2.12038.

Gelman, A. and Rubin, D.B., 1992. Inference from iterative simulation using multiple sequences. *Statistical Science*, 7(4), pp.457–472.